Daniele Battesimo Provenzano

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Employment

Forensic Department of the Italian Police Rome, Italy
Technical Inspector in Forensic Science and Lecturer *Sept. 2025 – present*

- Carrying out Scanning Electron Microscope (SEM) and microanalytical investigations within the GSR Analysis Division. Involved in an international ENFSI interlaboratory experimentation on GSR particles persistence.
- Teaching in training courses for forensic science trainees.

Italian Association for the Teaching of Physics (AIF) Various locations across Italy
Lecturer, Coach and Examiner *May 2019 – present*

- Preparing and delivering lectures, lab sessions, and problem-solving activities for students training for the Italian, European and International Physics Olympiads.
- Writing exam problems and contributing to national programs that promote physics education and support talented students across Italy.

Education

Scuola Normale Superiore Pisa, Italy
Ph.D. in Physics: Plasmonics and Metamaterials; ***ABD*** *Nov. 2020 – Sept. 2025*
Diploma (M.Sc.) in Physics; ***100/100 cum Laude*** *Oct. 2018 – Sept. 2020*

University of Pisa Pisa, Italy
M.Sc. in Condensed Matter Physics; ***110/110 cum Laude*** *Oct. 2018 – Oct. 2020*

Scuola Superiore di Catania Catania, Italy
Diploma (B.Sc.) in Physics; ***70/70 cum Laude*** *Oct. 2015 – Sept. 2018*

University of Catania Catania, Italy
B.Sc. in Physics; ***110/110 cum Laude*** *Oct. 2015 – July 2018*

Liceo Scientifico "Leonardo da Vinci" Reggio Calabria, Italy
High School Diploma; ***100/100 cum Laude*** *Sept. 2010 – June 2015*

Additional Courses

"GSR Analysis": Four-week intensive training program on the use of Scanning Electron Microscope (SEM) and the application of X-ray, Raman, and FTIR spectroscopies for the characterization of organic and inorganic gunshot residues (Oct. 2025).

School for Forensic Police Inspectors: Selected through a national open competitive examination and completed a six-month training program in forensic science during Ph.D. studies. The program included Italian and international law, as well as experimental techniques used by the *Forensic Department of the Italian Police*, with a specialization in ballistics and gunshot residue analysis. Successfully appointed as *Forensic Inspector* upon completion of the training. (June 2023 – Gen. 2024)

"24 ECTS for Teaching": Completed four prerequisite courses in Pedagogy, Psychology, Anthropology, and Teaching Methods, required for teacher qualification in Italy. (Summer 2022)

"English for Scientific Communication": Completed a course on effective scientific writing and presentation, with a focus on preparing research papers and delivering talks at international conferences. (Spring 2022)

Afternoon English Courses: Attended continuous English language courses at a British School in my hometown, focusing on grammar, writing, and conversation practice (4 hours per week). Earned Cambridge English certifications up to level C1. (Oct. 2008 – June 2015)

Teaching Experience

Forensic Department of the Italian Police Caserta, Italy

Lecturer

- **Training Courses for Forensic Trainees:** Prepared and delivered 19 hours of lectures and hands-on training sessions on Electron Microscopy, Energy-Dispersive Spectroscopy, and standard forensic procedures for Gunshot Residue analysis, offered to the trainees of the Forensic Inspector course. (Nov. 2025)

Scuola Normale Superiore Pisa, Italy

Lecturer, Teaching Assistant and Examiner

- "**Physics School in Pisa**": Prepared and gave 28 hours of lectures about topics in Classical and Modern Physics and led the relative problem-solving sessions, examined and graded 200 applicants per year through both written and oral exams. (2019 - 2025)
- "**University Orientation Courses**": Designed and delivered 12 hours of interactive lessons on Problem-Solving Techniques for 260 handpicked high school students, offering them a first taste of university-level physics and introducing everyday physics curiosities (June - July 2024)
- **"Classical Mechanics and Thermodynamics" course:** Served as a teaching assistant, delivering 30 hours of problem-solving sessions for first-year university students. (Fall 2020 - Spring 2021)

Italian Association for the Teaching of Physics Various locations across Italy

Lecturer, Coach and Examiner

- **"One step away from IPhO":** Prepared and gave 32 hours of lectures on Celestial Mechanics, Electromagnetism and Thermodynamics, aimed at the top 15 Italian students preparing to compete in the international Physics Olympiads. Selected the team members taking part in IPhO and EuPhO through theoretical and experimental examinations. (May 2019/23/24/25)
- **Physics Olympiad Summer School:** Led 16 hours of problem-solving sessions on Orbital Mechanics, Estimation Problems, Dimensional Analysis and Fluid Dynamics, aimed at 20 handpicked students taking part in the Physics Olympiad. (Aug. 2019/24/25)

Scuola Superiore di Catania Catania, Italy

Lecturer, Coach and Examiner

- **Training camp for the Physics Olympiad:** Prepared and gave 6 lectures about Electrostatics, Problem-Solving Techniques and Fluid Dynamics, aimed at 20 handpicked students taking part in the national finals of the Physics Olympiad. Led the relative problem-solving sessions, prepared and graded the final theoretical competition. (Mar. 2023/24/25)
- **Training camp for the Math Olympiad:** Led two problem-solving sessions focused on Combinatorics and Game Theory, aimed at 25 handpicked students taking part in the national finals of the Math Olympiad. (April 2016 and 2017)

Conference Talks and Posters

Metamaterials'2025, Amsterdam (NL), Sept. 2025

Graviton-Polaritons in FQH Systems Coupled to Optical Cavities (P)

GIREP-EPEC Conference in Physics Education, Leiden (NL), July 2025

When Assumptions Make the Difference: The Curious Case of the Mistreated Bernoulli's Equation (T)

META 2024, Toyama (JP), July 2024

Photonic Modes in Gyrotropic-Hyperbolic Heterostructures (P)

Publications

D. B. P., Z. Bacciconi, I. Carusotto, M. Dalmonte, G. C. la Rocca, *Graviton-Polaritons in FQH systems coupled to optical cavities*, in preparation.

D. B. P., *A comprehensive tutorial on Bernoulli's equation*, in preparation.

D. B. P., *The incredible amount of physics behind the Heron's fountain*, under review by the AJP.

D. B. P., *Debunking Boyle's self-flowing flask*, accepted by the EJP.

Stefania Carletti and D. B. P., *The Ninth Edition of the European Physics Olympiad*, chapter of *Speciale Olimpiadi 2025* (in Italian), pubblished by Associazione per l'Insegnamento della Fisica (AIF). (Coming in Dec. 2025)

D. B. P., Andrea Stefanini, Unblowing bubbles: understanding the physics of bubble deflation through a straw, Am. J. Phys. **93**, 797–805 (2025)

D. B. P. and Giuseppe C. la Rocca, Bulk and surface modes in a one-dimensional gyro-uniaxial photonic crystal, Phys. Rev. A **110**, 033514 (2024).

D. B. P. and Giuseppe C. la Rocca, Interface mode between gyroelectric and hyperbolic media, J. Opt. Soc. Am. B **40**, 172 (2023).

Awards & Achievements

Italy Team Leader at EuPhO 2025: Coached and led the Italian team at the European Physics Olympiad, held in Sofia, Bulgaria. (June 2025)

Best student of the *Forensic Ballistics Inspector* course: after six months of training, I ranked first in the final examinations among the ballistics inspector trainees and third overall among all the forensic science trainees, and was awarded a bronze medal. (Dec. 2023)

First Place in the National Open Competitive Exam for *Forensic Ballistic Inspectors*: Ranked first in the decennial national exam for ballistic experts to enroll in the Forensic Department of the Italian Police. (Apr. 2023)

Gold Medal – 53rd Rudolf Ortvay International Competition in Physics. (Feb. 2023)

Absolute Winner – 50th Rudolf Ortvay International Competition in Physics. (Dec. 2019)

SNS Scholarship for exceptional students: Awarded full scholarship (including tuition waiver, free housing, meals and a monthly stipend) after passing the selection process to enroll at Scuola Normale Superiore as a Master's student. (Oct. 2018 – Sept. 2020)

SSC Scholarship: Granted full scholarship (room and board) after passing the selection process to enroll at Scuola Superiore di Catania as an undergraduate student. (Oct. 2015 – Sept. 2018)

Projects Related to Physics Education

Italian Physics Olympiad (2019 – present) | *Italian website*

- Organized since 1987 by the *(Italian) Association for the Teaching of Physics* (AIF), the Italian Physics Olympiad selects and trains the country's top students for international competitions from among approximately thirty thousand participants each year. I joined the national training team in 2019. Since then, students I have trained have earned 2 gold, 16 silver and 28 bronze medals at IPhO and EuPhO. In recognition of my contributions, I was appointed to the national board responsible for preparing theoretical and experimental problems and for evaluating contestants at each stage of the competition.

Team Physics Competition (2023 – present) | *Italian website*

- The TPC is multi-stage event for Italian high school students that involves approximately three thousand participants each year, created by my university peers and me. My primary responsibility involves creating, refining and selecting the theoretical problems used in the competition.

Physics School in Pisa (2019 – present) | *Italian website*

- Since 2018, a team of physics students of Scuola Normale Superiore (SNS) has organized an annual one-week school for 24 top-performing Italian high school students. The program combines advanced problem solving sessions with an introduction to university-level physics through both classroom lectures and laboratory activities. In 2019, the project won the "Didactics of Physics" award from the Italian Physical Society. Since enrolling at SNS in 2019, my main duties have included evaluating and grading about 200 applicants per year through written and oral examinations, preparing original exam problems, and delivering lectures and problem-solving sessions.

Physics Camp in Catania (2023 – present)

- Since 2019, students of the Scuola Superiore di Catania have organized a physics training camp for participants from southern Italy selected for the national phase of the Italian Physics Olympiad. The program comprises theoretical lectures, problem-solving workshops, laboratory activities, and a concluding competition. I joined the organizing team in 2023.

Danny & Tony's Lab (2025 – present)

- Together with a computer scientist friend, I am developing a fully custom-built virtual and interactive physics laboratory hosted on a website. The platform will allow users to perform a variety of physics experiments within a 3D environment where they can walk around and interact with the apparatus, as in a video game. Some of these experiments are of my own design—several of which have been, or will be, presented in my educational publications—and cover topics in both classical and modern physics. One section of the lab will include an astronomical observatory for celestial mechanics simulations. The project aims to serve as an innovative and highly engaging tool for physics education.

"Problem-Solving Techniques in Physics, with Original Problems and Debunked Myths"

- This is a textbook on problem-solving techniques that I am writing in my free time, aimed at students preparing for the Physics Olympiad and first-year university students. It is specifically designed to debunk common myths and address misconceptions often found in traditional teaching and textbooks. I aim to complete it by 2027.

Certifications and Skills

Languages: English (C1, 195/210, certificated by Cambridge English), French (B1, certificated by Scuola Normale Superiore).

Major computer skills: MATLAB, Mathematica, Python, LaTeX, Git, HTML, video editing SWs.

Others: Fire Safety and First Aid (both certificated by the Italian Police).